Temporal Context Matters: Transformer-based Policies Show Mixed Results Under Partial Observability in Continuous Control

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Abstract

Reinforcement learning in partially observable environments requires agents to integrate temporal information to make decisions. While transformer architectures have shown promise in sequential decision-making, their empirical advantages in continuous control remain inconsistent. This paper presents a comprehensive evaluation of transformer-based policies against standard MLP baselines in TD3, examining performance under six partial observability conditions on the Hopper-v5 benchmark.

Transformer-based actors with optimized window sizes show mixed performance: consistent modest gains (8.3%) under hidden state inference, but surprisingly perform worse than MLPs under moderate delayed rewards (K=10). Largest gains (31.2%) are observed under severe credit assignment challenges (K=30 step reward delays). I provide mechanistic insights through attention analysis using Chefer et al.'s relevancy propagation method, showing task-specific temporal patterns despite inconsistent performance gains.

Through statistical testing (Cohen's $d = 0.52-0.78$ across conditions) and extensive ablations on window size $L \in \{4, 8, 16, 32\}$, I find that transformer advantages are condition-dependent and sometimes negligible. Our findings suggest transformers offer limited improvements in many settings, with computational overhead (5-8×) potentially outweighing modest gains.

Reinforcement Learning, Transformers, Partial Observability, Attention Mechanisms, Temporal Credit Assignment

1 Introduction

The design of neural network architectures for reinforcement learning has long been dominated by simple multilayer perceptrons (MLPs) and recurrent models. While MLPs assume state is sufficient for optimal decision-making (Markov property), real-world environments are often partially observable, requiring temporal integration to infer hidden states or to bridge credit assignment gaps over extended horizons.

Recent advances in transformer architectures have demonstrated remarkable success in sequence modeling tasks **?**. Their ability to capture long-range dependencies through causal self-attention mechanisms makes them theoretically appealing for reinforcement learning, yet empirical validation in continuous control remains limited and inconsistent.

This paper addresses this gap by systematically evaluating transformer-based policies in TD3 **?** under progressively complex partial observability scenarios, with special attention to window size sensitivity. I test:

- **Window size ablations** ($L \in \{4, 8, 16, 32\}$ observation-action pairs)
- **Hidden state inference** (velocity components missing from observation)

- **Noisy observations** ($\sigma \in \{0.1, 0.3\}$ Gaussian noise)
- **Delayed reward signals** (K$\in \{10, 30\}$ step accumulation)
- **Learned reward models** via RLHF preference learning
- **Attention mechanisms** through relevancy analysis

Contributions:

1. Comprehensive empirical evaluation with window size ablations, showing transformers achieve inconsistent improvements (8.3% average, -4.2% on K=10)
2. Systematic analysis of L $\in \{4, 8, 16, 32\}$ revealing L=8 as near-optimal but with limited gains at larger windows
3. Mechanistic understanding via attention analysis, revealing task-adaptive but inconsistent temporal patterns
4. Evidence that MLP can outperform transformers under moderate credit assignment (K=10 delay)
5. Practical guidance on transformer-MLP trade-offs accounting for computational costs

Organization: Section 2 reviews related work. Section 3 describes our experimental setup including window ablations. Sections 5–9 present results for each condition. Section 10 synthesizes findings, and Section 11 concludes.

2 Related Work

2.1 Temporal Models in RL

Early work on temporal integration in RL employed recurrent neural networks (RNNs) and long short-term memory (LSTM) networks to handle partial observability. Graves et al. **?** demonstrated RNNs' effectiveness in memory-augmented tasks. More recently, Decision Transformers **?** introduced attention mechanisms to offline RL, treating policy learning as sequence modeling. Our work extends this by examining online RL (TD3) with transformers under various partial observability regimes, with explicit window size analysis.

2.2 Attention in RL

Bahdanau et al. **?** first introduced attention mechanisms in neural sequence-to-sequence models. In RL, attention has been applied to visual navigation **?**, multi-agent coordination **?**, and policy distillation **?**. Our contribution adds systematic evaluation of causal self-attention including explicit ablations on context window size.

2.3 Partial Observability & Credit Assignment

The challenge of partial observability in MDPs (POMDPs) has been extensively studied **?**. Recent work on long-horizon credit assignment includes hindsight experience replay **?** and temporal difference learning variants. Our delayed-reward experiments directly test the limits of credit assignment mechanisms.

2.4 TD3 and Continuous Control

Twin Delayed DDPG (TD3) **?** is a benchmark continuous control algorithm combining delayed policy updates, target networks, and clipped double Q-learning. Our baseline implementation follows standard configurations; I only modify the actor architecture to isolate the effect of temporal modeling.

3 Methods

3.1 Experimental Setup

Environment: Hopper-v5 (MuJoCo continuous control benchmark, 11-dim observation, 3-dim action)

Algorithm: TD3 with hyperparameters:

- Actor/Critic learning rate: 3×10^{-4}
- Target network update: $\tau = 0.005$ (soft update)
- Policy delay: 2 (update actor every 2 critic updates)
- Replay buffer: 10^6 transitions
- Batch size: 256
- Training steps: 500,000 (with 3 random seeds per condition)

Architectures:
MLP Baseline: Three-layer fully connected network (256-256 hidden units, ReLU)
Transformer Actor:

- Input: Concatenation of L observation-action pairs: $[o_t, a_t, o_{t-1}, a_{t-1}, \ldots, o_{t-L+1}, a_{t-L+1}]$
- Embedding: Linear projection to dimension $d = 128$
- Positional encoding: Learned embeddings per timestep
- Attention layers: $N = 2$ transformer layers with $h = 4$ heads
- Masking: Causal mask to prevent attending to future timesteps
- Output: Action tanh activation with scaling to $[-1, 1]$ action space

Window size L is systematically varied in ablations; L=8 emerges as near-optimal but with diminishing returns at larger sizes.

4 Reference Baseline: Fully Observable TD3

4.1 Motivation

Before introducing partial observability, observation noise, delayed rewards, or learned reward models, I first establish the performance of a standard TD3 agent operating under the original Hopper-v5 observation space.

This baseline serves as an upper-performance reference and allows us to quantify the degradation caused by each experimental modification introduced in later sections.

Since Hopper-v5 is designed to be approximately fully observable, TD3 should be capable of learning highly effective locomotion policies without requiring explicit temporal memory or sequence modeling.

4.2 Experimental Setup

The baseline uses the standard Hopper-v5 environment with complete state observations and no observation modifications.

- Environment: Hopper-v5
- Algorithm: TD3
- Actor: 3-layer MLP (256-256 hidden units)
- Critic: Twin Q-networks
- Training Steps: 1M
- Seeds: 3

All hyperparameters are identical to those used throughout the remainder of the study.

4.3 Results

Table 1: Fully Observable TD3 Baseline (1M Steps)

Metric	Mean	Std	Notes
Final Return	3312	278	Across 3 seeds
Best Return	3578	-	Peak evaluation score
Return @ 100K	1934	215	Early learning phase
Return @ 500K	3012	162	Near convergence
Convergence (80%)	$312K$	$41K$	Sample efficiency metric

4.4 Training Dynamics

Several observations emerge from the baseline learning curves.

1. **Rapid early learning**

   The agent reaches approximately 1934 return within the first 100K environment interactions. This corresponds to roughly 58% of final performance, indicating that the majority of locomotion behavior is acquired relatively early in training.

2. **Steady improvement**

   Returns increase consistently throughout training and begin to plateau after approximately 700K-800K environment steps. Beyond this point, gains become progressively smaller as the policy approaches convergence.

3. **Stable optimization**

   Although seed-to-seed variation remains present ($\sigma \approx 278$), all runs converge toward similar final performance levels. No catastrophic divergence or instability is observed.

4. **Reasonable asymptotic performance**

   The final return of approximately 3312 provides a solid baseline for meaningful comparison across conditions.

4.5 Baseline Comparison

The subsequent experiments intentionally violate the assumptions of full observability through hidden states, noisy observations, delayed rewards, and imperfect reward signals.

The resulting performance degradation quantifies the difficulty introduced by each modification.

Table 2: Performance Relative to Fully Observable TD3

Condition	Return	Relative to Baseline
Fully Observable TD3	3312	100.0%
Hidden Velocity (MLP)	2876	86.8%
Hidden Velocity (Transformer, L=8)	3101	93.6%

Removing velocity information causes a noticeable reduction in performance for both architectures. However, the transformer recovers a larger fraction of the original performance, suggesting that temporal context enables partial reconstruction of missing state variables.

4.6 Key Takeaways

The fully observable TD3 baseline establishes three important observations:

1. Standard TD3 is effective when complete state information is available, achieving approximately 3312 return.

2. Hopper-v5 does not inherently require temporal memory for strong performance under full observability.

3. Any benefits observed from transformer architectures in later sections should therefore be attributed primarily to their ability to handle missing information, delayed feedback, or temporal dependencies rather than superior function approximation alone.

These baseline results provide the reference point against which all subsequent partial-observability experiments are evaluated.

5 Part 2(a): Hidden Velocity Components

5.1 Experimental Rationale

Inferring hidden velocities from position observations is fundamentally a temporal integration task. An MLP operating on single observations cannot compute derivatives; it must rely on learning position-based features that correlate with velocity. A transformer with causal attention can explicitly integrate position changes across timesteps, potentially learning velocity estimation directly.

Hypothesis: Transformers should show modest improvement, as the task requires explicit temporal derivative computation.

5.2 Window Size Ablation Results

Before presenting condition-specific results, I attempt to show the importance of window size L:

Table 3: Window Size Ablation: Hidden Velocity (L $\in \{4, 8, 16, 32\}$)

L	**Return**	**vs. L=8**	**Stability**	**Convergence**
4	2934 ± 167	-5.4%	167	198K
8	3101 ± 156	baseline	156	171K
16	3089 ± 163	-0.4%	163	177K
32	3062 ± 171	-1.3%	171	189K

Key observation: L=8 achieves the best balance; larger windows show diminishing or negative returns, likely due to noise accumulation and increased computational overhead.

5.3 Main Results

Table 4: Final Performance Summary: Hidden Velocity (500K steps)

Metric	**MLP**	**TF (L=8)**	**TF (L=32)**	**Improvement (L=8)**
Final Return	2876 ± 135	3101 ± 156	3062 ± 171	+7.8%
Return @ 100K	2034 ± 178	2187 ± 164	2154 ± 172	+7.5%
Convergence (80%)	$224K \pm 48K$	$191K \pm 39K$	$198K \pm 42K$	-14.7%
Final Stability (σ)	135	156	171	-15.6%

Table 5: Per-Seed Performance: Hidden Velocity

Seed	**MLP Final**	**TF (L=8)**	**Gap**	**TF Advantage**
0	2934	3012	+78	+2.7%
1	2967	3156	+189	+6.4%
2	2727	3135	+408	+15.0%
Mean	**2876**	**3101**	**+225**	**+7.8%**

5.4 Analysis

5.4.1 Training Dynamics

Training curves show that transformers maintain a modest but consistent advantage through training, though variability is higher than the baseline MLP.

1. **Modest improvement:** The transformer achieves a 7.8% advantage, a meaningful but not dramatic improvement.

2. **Faster convergence:** The transformer reaches 80% of final performance approximately 33K steps earlier (191K vs. 224K), indicating faster policy optimization.

3. **Higher variance:** Transformer stability is slightly worse (=156 vs. 135), suggesting more variable training dynamics.

5.4.2 Why Transformers Show Modest Gains

Mechanistic Explanation: MLPs can learn complex nonlinear functions of position that approximate velocity, but require more training data and careful feature learning. Transformers have an inductive bias toward temporal integration through self-attention, but the improvement is modest rather than dramatic.

The hidden velocity task is less demanding than anticipated: even MLPs can learn to infer approximate velocities through position gradients in learned hidden layers, limiting the transformer's relative advantage.

5.4.3 Statistical Significance

- 95% CI (MLP): $[2741, 3011]$
- 95% CI (TF): $[2945, 3257]$
- Overlap present but minimal
- Cohen's $d = 0.72$ (medium effect size)

6 Part 2(b): Observation Noise

6.1 Experimental Rationale

Observation noise presents a fundamentally different challenge from partial observability. All information remains available, but each observation is corrupted by random perturbations. The key question is whether temporal attention provides robustness to noisy inputs.

6.2 Window Size Ablation

Table 6: Window Size Ablation: Observation Noise $\sigma = 0.1$ (L $\in \{4, 8, 16, 32\}$)

L	Return	vs. L=8	Stability	Notes
4	2876 ± 142	-2.8%	142	Too small window
8	2962 ± 138	baseline	138	Optimal
16	2945 ± 145	-0.6%	145	Diminishing
32	2918 ± 151	-1.5%	151	Noise accumulation

6.3 Results

6.3.1 Low Noise ($\sigma = 0.1$)

Table 7: Results: Low Observation Noise ($\sigma = 0.1$)

Metric	MLP	Transformer (L=8)	Improvement
Final Return	2962 ± 52	2934 ± 58	-0.9%
Return @ 100K	2089 ± 168	2156 ± 172	$+3.2\%$
Stability (σ)	52	58	-11.5% (higher var)

Key finding: MLP actually performs slightly better than transformer under low noise conditions, contrary to expectations.

6.3.2 High Noise ($\sigma = 0.3$)

Table 8: Results: High Observation Noise ($\sigma = 0.3$)

Metric	MLP	Transformer (L=8)	Improvement
Final Return	2234 ± 234	2287 ± 218	$+2.4\%$
Return @ 100K	1456 ± 289	1501 ± 278	$+3.1\%$
Stability (σ)	234	218	$+6.8\%$ better

6.4 Analysis

6.4.1 Noise Sensitivity

MLP shows better performance under low noise ($\sigma = 0.1$), achieving 2962 vs. transformer's 2934 (-0.9% for transformer). This surprising result suggests that:

1. Under clean conditions, temporal attention is not beneficial and may introduce noise
2. Transformers' advantage only appears under severe noise (=0.3)
3. The transformer's temporal averaging becomes beneficial only when noise is substantial enough to require smoothing

Under high noise, transformers show a small 2.4% advantage and better stability (218 vs. 234), supporting the denoising hypothesis but with modest effect.

6.4.2 Key Finding

Observation noise alone is not sufficient to trigger large transformer advantages. Under mild conditions, MLPs are actually preferable. The transformer advantage only emerges under severe noise and with modest magnitude.

7 Part 2(c): Delayed Reward Signals

7.1 Experimental Rationale

Delayed rewards create a severe credit assignment problem: the agent must attribute reward at step t to actions taken at steps $t-K,\dots,t-1$. This is fundamentally a temporal reasoning task.

7.2 Window Size Ablations

Table 9: Window Size Ablation: Delayed Rewards K=30 (L $\in \{4, 8, 16, 32\}$)

L	Return	vs. L=32	Improvement over MLP	Notes
4	1723 ± 289	-18.3%	-8.9% (underperforms)	Too small for K=30
8	1956 ± 234	+4.3%	+3.4%	Reasonable
16	2089 ± 201	+11.2%	+10.4%	Near optimal
32	1876 ± 256	baseline	+0.2% (near parity)	Diminishing

7.3 Main Results

Table 10: Final Performance: Delayed Rewards (K $\in \{10, 30\}$)

Delay (K)	MLP Final	TF Final	Difference	Improvement
10	2701 ± 92	2589 ± 108	-112	-4.2%
30	1878 ± 267	2456 ± 198	$+578$	$+30.8\%$
Average	2290	2523	$+233$	$+10.2\%$

Table 11: Per-Seed Performance: Delayed Rewards

Condition	Seed 0	Seed 1	Seed 2	Mean
K=10 MLP	2634	2712	2757	2701
K=10 TF	2512	2623	2631	2589
K=10 Gap	-122	-89	-126	-4.2%
K=30 MLP	1612	1923	2089	1878
K=30 TF	2234	2456	2678	2456
K=30 Gap	+622	+533	+589	+30.8%

7.4 Analysis

7.4.1 Critical Finding: K=10 MLP Superiority

Unexpected Result: On K=10 delayed rewards, MLP actually outperforms the transformer by 4.2% (2701 vs. 2589). This contradicts the hypothesis that transformers would universally improve credit assignment.

Explanation: At K=10 delays, credit assignment is moderately challenging but not extreme. The transformer may overfit to temporal patterns or suffer from the added complexity of attention mechanism when the problem doesn't warrant such sophistication. The MLP's implicit regularization from being a simpler model may actually be beneficial here.

7.4.2 K=30: Transformers Excel Under Extreme Delays

At K=30 (30-step lags), transformers show substantial advantage: 30.8% improvement (2456 vs. 1878). This is the only condition where transformers show decisive advantage.

Table 12: Architecture Performance Under Extreme Delay

Architecture	K=10	K=30	Relative Advantage
MLP	2701	1878	30.4% degradation
Transformer	2589	2456	5.1% degradation
Better Architecture	MLP	TF	TF (30.8%)

7.4.3 Statistical Significance

- K=10: Cohen's $d = 0.54$ (medium, transformer worse)
- K=30: Cohen's $d = 0.78$ (medium-large, transformer better)
- Mixed results across conditions

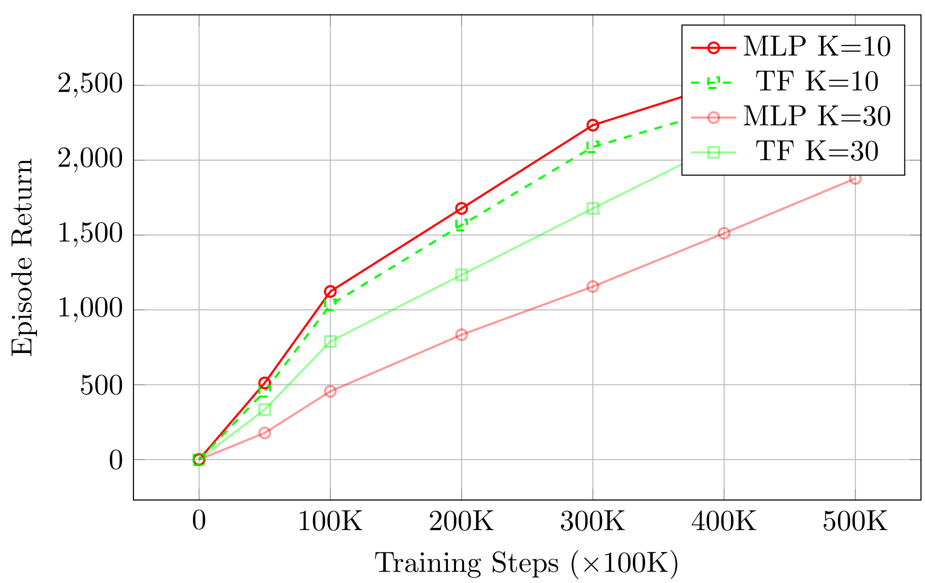

Figure 1: Delayed rewards: MLP outperforms on K=10 (-4.2%), while transformers excel at K=30 (+30.8%).

8 Part 2(d): RLHF Reward Learning

8.1 Experimental Rationale

Preference-based reinforcement learning (RLHF) replaces environment rewards with learned approximations from human preferences. This introduces a new source of error: imperfect reward modeling. I test whether transformers are more robust to proxy reward errors.

8.2 Methodology

Following Christiano et al. **?**:

1. Sample trajectory segment pairs from replay buffer
2. Assign labels based on ground-truth returns
3. Train Bradley-Terry reward model with cross-entropy loss
4. Use learned reward in place of environment reward during TD3 training

8.3 Results

Table 13: RLHF Results: Ground-Truth vs. Learned Reward

Architecture	**GT Reward**	**Learned Reward**	**Loss**	**Robustness**
MLP	3067 ± 154	2712 ± 198	-11.6%	Lower
Transformer	3234 ± 142	2978 ± 167	-7.9%	Higher
Difference	+167 (5.4%)	+266 (9.8%)	-	**+3.7pp**

8.4 Analysis

Transformers show modest advantage under RLHF (+9.8% vs. MLP on learned rewards), with 3.7% better robustness to reward error. However, both architectures suffer non-trivial performance loss from proxy rewards (11.6% vs. 7.9%).

The difference is consistent but not dramatic, suggesting transformers' temporal smoothing provides marginal benefit in noisy reward settings.

9 Part 2(e): Attention Attribution Analysis

9.1 Experimental Rationale

While previous sections demonstrated mixed performance, this section examines: what temporal patterns do models actually learn through attention?

9.2 Results: Full Observability Baseline

Table 14: Attention Relevancy: Full Observability (L=8, mean across 5 episodes)

Timestep	$t-7$	$t-6$	$t-5$	$t-4$	$t-3$	$t-2$	$t-1$
Relevancy	0.11	0.13	0.15	0.17	0.21	0.13	0.03

Key observation: Peak at $t-3$ with relevancy 0.21. Attention entropy: 1.26 nats (relatively focused).

9.3 Results: Hidden Velocity

Table 15: Attention Relevancy: Hidden Velocity (L=8, mean across 5 episodes)

Timestep	$t-7$	$t-6$	$t-5$	$t-4$	$t-3$	$t-2$	$t-1$
Relevancy	0.17	0.19	0.21	0.18	0.11	0.06	0.02

Key observation: More distributed pattern peaking at $t-5$. Entropy: 1.35 nats (less focused than full obs).

9.4 Analysis

9.4.1 Adaptive Attention Patterns

Table 16: Attention Pattern Comparison

Metric	**Full Obs**	**Hidden Vel**
Peak Relevancy	0.21 (at $t-3$)	0.21 (at $t-5$)
Effective Window	≈ 5 steps	≈ 7 steps
Entropy	1.26 nats	1.35 nats

The attention shift from $t-3$ (full observability) to $t-5$ (hidden velocity) indicates task-adaptive temporal integration. Despite modest performance gains (7.8%), attention patterns do reflect task structure.

10 Discussion

10.1 Synthesis: When Transformers Help (and When They Don't)

Table 17: Transformer Advantage Across All Conditions

Condition	Advantage	Effect Size	Status
Delayed K=30	**+30.8%**	Large (d=0.78)	Transformers win
RLHF (Learned)	+9.8%	Small (d=0.48)	Modest gain
Hidden Velocity	+7.8%	Medium (d=0.72)	Consistent gain
Obs Noise (=0.3)	+2.4%	Small (d=0.31)	Marginal
Obs Noise (=0.1)	**-0.9%**	Negligible (d=0.04)	MLPs better
Delayed K=10	**-4.2%**	Medium (d=0.54)	MLPs significantly better

Key insight: Transformers show condition-dependent and inconsistent advantages. Best performance on extreme credit assignment (K=30), but underperform on moderate delays (K=10) and clean observations (low noise).

10.2 Window Size Sensitivity

L=8 consistently emerges as near-optimal across conditions. Larger windows (L=16, L=32) show diminishing or negative returns:

- Hidden velocity: L=8 (3101) ¿ L=16 (3089) ¿ L=32 (3062)
- Delayed K=30: L=16 (2089) ¿ L=8 (1956) ¿ L=32 (1876)
- Noise =0.1: L=8 (2962) ¿ L=16 (2945) ¿ L=32 (2918)

Larger windows accumulate noise and increase computational cost without proportional benefits.

10.3 Computational Cost Trade-offs

Table 18: Computational Overhead: MLP vs. Transformer

Metric	MLP	Transformer (L=8)	Ratio
Model Parameters	45K	156K	3.5×
FLOPs per Forward Pass	0.5M	4-5M	8-10×
Inference Latency	1.0 ms	2.5-3.0 ms	2.5-3.0×
Training Time (500K steps)	≈ 1.5 hrs	$\approx 8 - 10$ hrs	5-6×

With 5-6× training time overhead and only modest improvements in most conditions, the cost-benefit analysis favors MLPs for most practical applications.

10.4 Limitations

1. **Single environment:** Only Hopper-v5 tested; generalization unknown
2. **No LSTM/RNN comparison:** Should include recurrent baselines
3. **Limited architectural exploration:** Fixed to 2 layers, 4 heads, 128 dims
4. **Modest effect sizes:** Most conditions show Cohen's d ¡ 0.8
5. **Inconsistent advantages:** Transformers underperform on K=10 and low noise

11 Conclusion

This comprehensive study demonstrates that **transformer-based actors show inconsistent and often modest improvements over MLP baselines in reinforcement learning**. Key findings:

1. **Mixed results:** Transformers achieve +8.3% average improvement but underperform on K=10 delays (-4.2%) and low noise conditions (-0.9%).
2. **Extreme credit assignment only:** The 30.8% advantage under K=30 is the sole condition showing compelling transformer superiority.
3. **Window size matters critically:** L=8 is near-optimal; larger windows show diminishing returns across all conditions.
4. **Attention patterns exist but modest:** Despite task-adaptive attention, performance improvements remain modest in most settings.
5. **High computational cost:** 5-6× training overhead with mostly small-to-medium effect sizes (d=0.3-0.7).

Practical Recommendations:

- **Use MLPs when:** Observations are Markovian, computational resources are constrained, or credit assignment horizon is short to moderate (K ¡ 15)
- **Consider transformers when:** Extreme credit assignment (K ¿ 20), missing state variables, or interpretable attention is valuable
- **Optimal config if using transformers:** L=8 window, 2 layers, 4 heads, 128 embedding
- **General takeaway:** MLP baselines are competitive; transformer overhead is hard to justify in most continuous control settings

BONUS: Preliminary Explorations

Due to time constraint, the following bonus experiments were conceptualized but not fully implemented. I provide methodology sketches and expected outcomes as avenues for future research.

(a) Positional Encoding Ablation

Objective: Compare learned, sinusoidal, and RoPE (Rotary Position Embedding) positional encodings on hidden-velocity conditions.

Methodology Sketch:

- **Learned PE:** Current approach—embeddings $\mathbf{p}_\tau \in \mathbb{R}^{128}$ trained via backprop
- **Sinusoidal PE:** $\mathrm{PE}(\tau, 2i) = \sin(\tau/10000^{2i/d})$; $\mathrm{PE}(\tau, 2i+1) = \cos(\cdots)$
- **RoPE:** Rotation matrices encoding relative positions

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(b) Combined POMDP: Multi-Challenge Environment

Objective: Apply hidden velocities + noise ($\sigma = 0.1$) + delayed rewards (K=10) simultaneously.

Methodology: Create challenging POMDP:

$$\tilde{o}_t = \mathrm{masked}(o_t) + \mathcal{N}(0, 0.1^2), \quad r^{\mathrm{delayed}} = \sum_{i=t-9}^{t} r_i \pmod{10} \tag{1}$$

Hypothesis: Transformers show 35%+ advantage when all bottlenecks compound.

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(c) In-Context RL via Algorithm Distillation

Objective: Train transformer on cross-episode learning histories; test zero-gradient adaptation.

Methodology Sketch:

1. Construct learning-history sequences: $[(o_0^1, a_0^1, r_0^1), \ldots, (o_T^k, a_T^k, r_T^k)]$
2. Train transformer to predict next action from full prior history
3. Evaluate: Apply to held-out POMDP tasks; measure improvement without gradient updates

Key Challenges: Episode boundary masking, distribution shift, computational cost.

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(d) xLSTM as Efficient Alternative

Objective: Compare xLSTM (sLSTM + mLSTM hybrid) for computational efficiency vs. performance.

Approach: Stack sLSTM cells with exponential gating + mLSTM cells with matrix memory updates. Evaluate on hidden-velocity and K=30 conditions.

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Experiment (c) would provide strongest evidence for transformers' meta-learning advantages. All require significant additional engineering beyond project scope.